Assignment No :2

Name : Hidayat Ullah

Registration No:BF25NWELE0685

Section : B

Department :Electrical (AI)

Submitted to: Rizwan sir

Subject Title: COMPUTER PROGRAMMING

Program 1:

Write a program to: • Take 10 integer inputs in an array. • Find and display: Maximum value Minimum value Average

```
#include <iostream>   // Library for input/output
using namespace std;  // Use standard namespace

int main() {
    int arr[10];      // Array to store 10 integers
    int sum = 0;      // Variable to store sum

    cout << "Enter 10 integers:\n";

    // Loop to take input
    for(int i = 0; i < 10; i++) {
        cin >> arr[i];      // Store value in array
        sum += arr[i];      // Add value to sum
    }

    int max = arr[0];  // Assume first value is max
    int min = arr[0];  // Assume first value is min

    // Loop to find max and min
    for(int i = 1; i < 10; i++) {
        if(arr[i] > max)   // If current value is greater
            max = arr[i];  // Update max

        if(arr[i] < min)   // If current value is smaller
            min = arr[i];  // Update min
```

```
    }

    float avg = sum / 10.0;  // Calculate average

    cout << "Maximum = " << max << endl;
    cout << "Minimum = " << min << endl;
    cout << "Average = " << avg << endl;

    return 0;  // End program
}
```

✎ Program 2:

Write a program that: • Uses a function to reverse a 1D array. • Display the array before and after reversal.

```
#include <iostream>
using namespace std;

// Function to reverse array
void reverseArray(int arr[], int size) {
    // Loop runs till half of array
    for(int i = 0; i < size / 2; i++) {
        // Swap first with last
        swap(arr[i], arr[size - i - 1]);
    }
}

int main() {
    int arr[5] = {1, 2, 3, 4, 5};  // Initialize array

    cout << "Before Reverse:\n";

    // Display original array
    for(int i = 0; i < 5; i++)
        cout << arr[i] << " ";

    reverseArray(arr, 5);  // Call function

    cout << "\nAfter Reverse:\n";

    // Display reversed array
    for(int i = 0; i < 5; i++)
        cout << arr[i] << " ";

    return 0;
}
```

➮ Program 3:

Write a program to: • Input two 3 × 3matrices. • Perform matrix addition. • Display the result matrix.

```
   #include <iostream>        // Include library
using namespace std;       // Use standard
namespace

int main() {                          // Main function
   int A[3][3], B[3][3], C[3][3];    // Declare three
3x3 matrices

   cout << "Enter first matrix:\n";  // Ask user for
first matrix

   for(int i = 0; i < 3; i++)        // Outer loop for
rows
      for(int j = 0; j < 3; j++)    // Inner loop for
columns
         cin >> A[i][j];            // Input value for
matrix A

   cout << "Enter second matrix:\n"; // Ask for
second matrix

   for(int i = 0; i < 3; i++)        // Loop rows
      for(int j = 0; j < 3; j++)    // Loop columns
         cin >> B[i][j];            // Input matrix B

   for(int i = 0; i < 3; i++)        // Loop rows
      for(int j = 0; j < 3; j++)    // Loop columns
         C[i][j] = A[i][j] + B[i][j];  // Add matrices

   cout << "Result Matrix:\n";       // Print result
message

   for(int i = 0; i < 3; i++) {     // Loop rows
      for(int j = 0; j < 3; j++)   // Loop columns
         cout << C[i][j] << " ";  // Print each element
      cout << endl;               // Move to next line
   }

   return 0;   // End program
}
```

Program 4: Write a program that: • Stores names of 5 students in a string array. • Sorts them in alphabetical order. • Displays the sorted list.

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```
#include <iostream>          // Include input-
output library
#include <string>            // Include string
library
using namespace std;         // Use standard
namespace

int main() {                 // Main function starts
   string names[5];          // Declare array of 5
strings

   cout << "Enter 5 names:\n";  // Ask user for
input

   for(int i = 0; i < 5; i++) { // Loop to input
names
      cin >> names[i];          // Store each name
   }

   for(int i = 0; i < 5; i++) {        // Outer loop
for sorting
      for(int j = i + 1; j < 5; j++) {// Inner loop
for comparison
         if(names[i] > names[j])     // Compare
strings alphabetically
            swap(names[i], names[j]); // Swap if
out of order
      }
   }

   cout << "Sorted Names:\n";   // Print heading

   for(int i = 0; i < 5; i++) { // Loop to display
names
      cout << names[i] << endl; // Print each
name
   }

   return 0;   // End program
}
```

- **Program 5:**

Power System Load Analysis (1D Array + Function) In a power system, hourly load demand (in MW) is recorded. Write a program to: • Store load data for 24 hours in an array. • Create functions to calculate: Total load Peak

```
#include <iostream>        // Include library
using namespace std;       // Use standard
namespace

int totalLoad(int arr[], int size) { // Function to
calculate total
   int sum = 0;                    // Initialize sum
   for(int i = 0; i < size; i++) { // Loop through
array
      sum = sum + arr[i];          // Add each value
   }
   return sum;                     // Return total
}

int peakLoad(int arr[], int size) { // Function to
find max value
   int max = arr[0];               // Assume first
value is max
   for(int i = 1; i < size; i++) {// Loop from
second element
      if(arr[i] > max)             // Check if greater
         max = arr[i];             // Update max
   }
   return max;                     // Return peak value
}

int main() {                       // Main function
   int load[24];                   // Array for 24 hours

   cout << "Enter 24 load values:\n"; // Ask for
input

   for(int i = 0; i < 24; i++) { // Loop for input
      cin >> load[i];              // Store each value
   }

   cout << "Total Load = " << totalLoad(load,
24) << endl; // Call function
   cout << "Peak Load = " << peakLoad(load,
24) << endl;   // Call function

   return 0;   // End program
}
```

- **Program 6:**

Design a program to: Store student names (string array) and their marks (1D array). Use functions to: Find topper Display pass/fail status Search a student by name

```
#include <iostream>        // Include library
#include <string>          // Include string
using namespace std;       // Use namespace

int findTopper(int marks[], int size) { // Function to find
topper
   int index = 0;                  // Assume first student is topper
   for(int i = 1; i < size; i++) {    // Loop through marks
      if(marks[i] > marks[index])    // Compare marks
         index = i;                // Update index
   }
   return index;                   // Return topper index
}

void searchStudent(string names[], int marks[], int size,
string key) { // Search function
   for(int i = 0; i < size; i++) {   // Loop through names
      if(names[i] == key) {         // If name matches
         cout << "Found: " << names[i] << " Marks: " <<
marks[i] << endl;
         return;                   // Exit function
      }
   }
   cout << "Student not found\n";   // If not found
}

int main() {                       // Main function
   string names[5];                // Array for names
   int marks[5];                   // Array for marks

   cout << "Enter names and marks:\n"; // Input message

   for(int i = 0; i < 5; i++) {     // Loop for input
      cin >> names[i] >> marks[i]; // Store name and marks
   }

   int topper = findTopper(marks, 5); // Call topper function

   cout << "Topper: " << names[topper] << endl; // Display
topper

   for(int i = 0; i < 5; i++) {     // Loop for pass/fail
      if(marks[i] >= 50)           // Check condition
         cout << names[i] << " Pass\n";
      else
         cout << names[i] << " Fail\n";
```

```
    }

    string key;                    // Variable for search
    cout << "Enter name to search: ";
    cin >> key;                    // Input search name

    searchStudent(names, marks, 5, key); // Call search
function

    return 0;   // End program
}
```

Program 7:

Renewable Energy Monitoring System (1D Array + Functions) A solar power plant records the energy generated (in kWh) every hour for one day.

Write a C++ program to:

- **Store energy generation data for 24 hours in a 1D array.**

Create functions to:

- **Calculate total energy generated in a day**
- **Find the hour with maximum generation**
- **Calculate average energy generation**
- **Display all results clearly.**

```
#include <iostream>       // Include library
using namespace std;      // Use namespace

float totalEnergy(float arr[], int size) { // Total energy
function
   float sum = 0;                    // Initialize sum
   for(int i = 0; i < size; i++) {      // Loop
      sum += arr[i];                 // Add values
   }
   return sum;                       // Return sum
}

int maxHour(float arr[], int size) {      // Function to
find max hour
   int index = 0;                    // Assume first index
   for(int i = 1; i < size; i++) {      // Loop
      if(arr[i] > arr[index])         // Compare
         index = i;                 // Update index
   }
   return index;                     // Return index
}

float average(float arr[], int size) {      // Average
function
```

```
    return totalEnergy(arr, size) / size; // Return average
}

int main() {                        // Main function
    float energy[24];               // Array of 24 hours

    cout << "Enter 24 values:\n";       // Input message

    for(int i = 0; i < 24; i++) {       // Loop
        cin >> energy[i];               // Store values
    }

    cout << "Total = " << totalEnergy(energy, 24) << endl;
    cout << "Peak Hour = " << maxHour(energy, 24) << endl;
    cout << "Average = " << average(energy, 24) << endl;

    return 0;   // End
}
```

program 8:

Battery Health Monitoring (1D Array + Functions) In battery management systems, voltage readings are monitored to ensure safe operation.

Write a program to:

• Store 20 voltage readings of a battery in an array. •

Define safe voltage limits (e.g., 3.0V to 4.2V).

• Create functions to:

• Count how many readings are below or above safe limits •

Find the maximum and minimum voltage •

Display a warning message if unsafe readings exist

```
#include <iostream>       // Include library
using namespace std;      // Use namespace

int main() {              // Main function
   float v[20];           // Array for 20 readings
   float low = 3.0, high = 4.2; // Safe limits

   int below = 0, above = 0; // Counters

   cout << "Enter 20 readings:\n";

   for(int i = 0; i < 20; i++) { // Input loop
      cin >> v[i];            // Store value

      if(v[i] < low)          // Check below limit
         below++;             // Increment
      else if(v[i] > high)    // Check above limit
         above++;             // Increment
   }

   float max = v[0], min = v[0]; // Initialize max and min

   for(int i = 1; i < 20; i++) { // Loop
      if(v[i] > max) max = v[i]; // Update max
      if(v[i] < min) min = v[i]; // Update min
   }

   cout << "Below = " << below << endl;
   cout << "Above = " << above << endl;
   cout << "Max = " << max << endl;
   cout << "Min = " << min << endl;

   return 0;   // End
}
```

program 9:

Smart Grid Load Distribution (2D Array + Functions) In a smart grid, electricity consumption is recorded across multiple regions.

Write a program to:

- **Store power consumption (in MW) for 4 regions over 7 days using a 2D array.**
- **Create functions to:**
- **Calculate total consumption for each region**
- **Identify the day with highest total demand**

• **Compute overall average consumption**

```
#include <iostream>       // Include library
using namespace std;      // Use namespace

int main() {              // Main function
   int arr[4][7];         // 2D array (4 regions, 7 days)

   cout << "Enter values:\n";

   for(int i = 0; i < 4; i++) {      // Loop rows
      for(int j = 0; j < 7; j++) {  // Loop columns
         cin >> arr[i][j];         // Input value
      }
   }

   for(int i = 0; i < 4; i++) {      // Loop regions
      int sum = 0;                   // Reset sum
      for(int j = 0; j < 7; j++) {  // Loop days
         sum += arr[i][j];          // Add values
      }
      cout << "Region " << i+1 << ": " << sum << endl;
   }

   return 0;   // End
}
```

program 10:

Classroom Attendance System (2D Array + Functions) Teachers often track student attendance daily.

Write a program to:

• **Store attendance for 5 students over 7 days using a 2D array (Use 1 for present, 0 for absent).**

• **Create functions to:**

• **Calculate total attendance of each student**

• **Find the student with highest attendance**

• **Calculate overall class attendance percentage**

```
#include <iostream>       // Include library
using namespace std;      // Use namespace

int main() {              // Main function
   int arr[5][7];         // 2D array (5 students, 7 days)

   cout << "Enter attendance (1/0):\n";

   for(int i = 0; i < 5; i++) {      // Loop students
      for(int j = 0; j < 7; j++) {  // Loop days
         cin >> arr[i][j];         // Input attendance
      }
   }

   for(int i = 0; i < 5; i++) {      // Loop students
      int sum = 0;                  // Reset total
      for(int j = 0; j < 7; j++) {  // Loop days
         sum += arr[i][j];          // Count present days
      }
      cout << "Student " << i+1 << ": " << sum << endl;
   }

   return 0;   // End
}
```